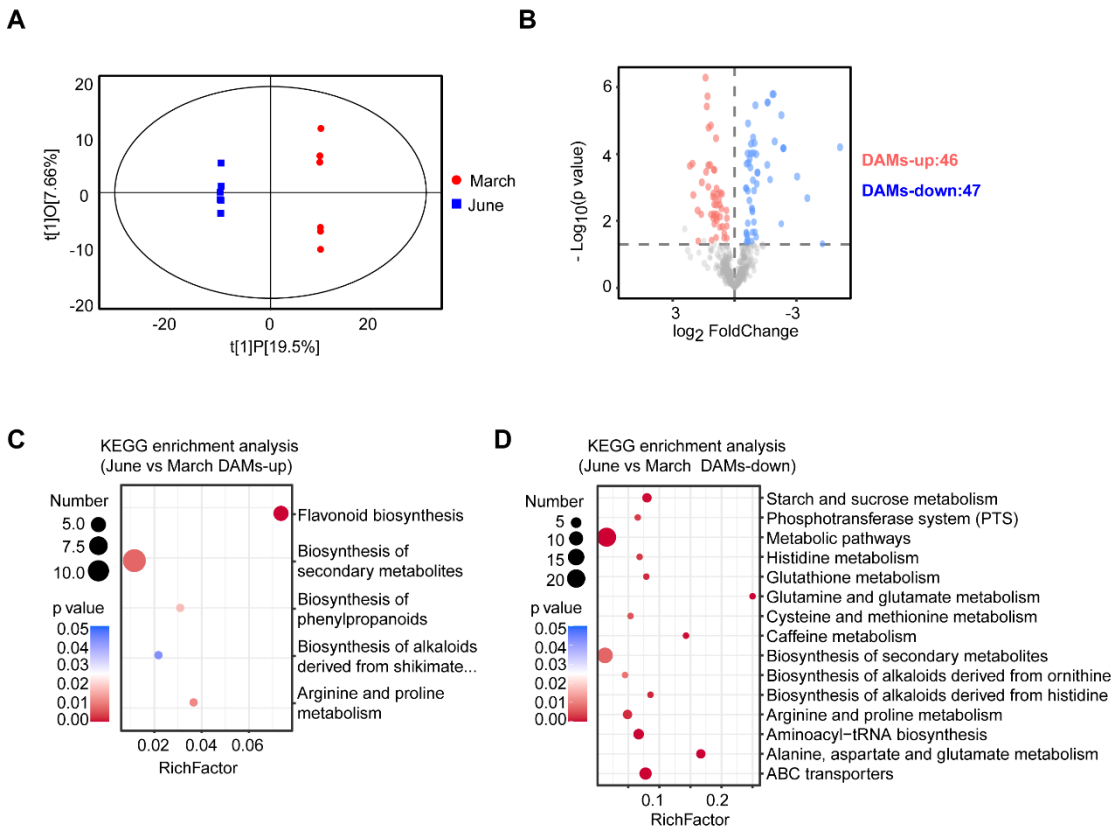
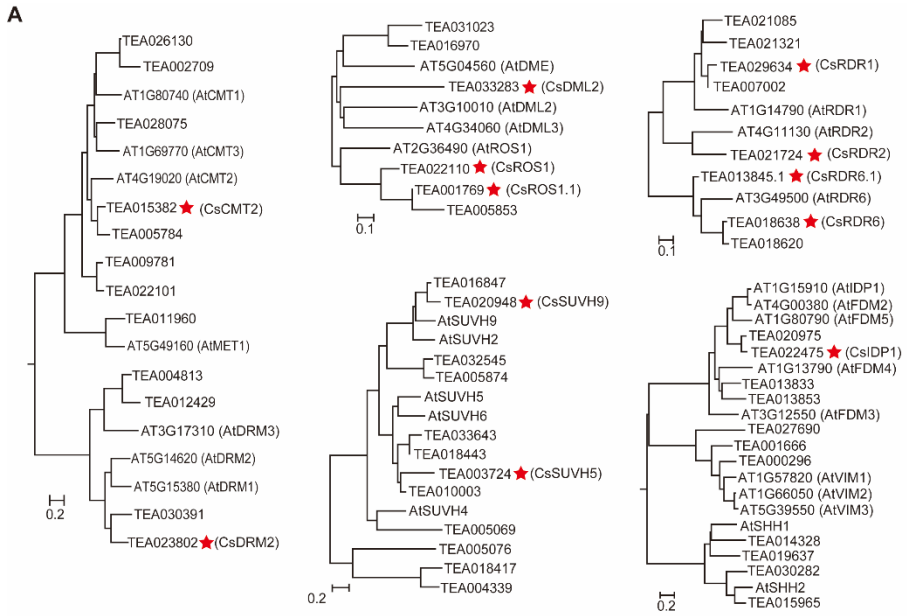
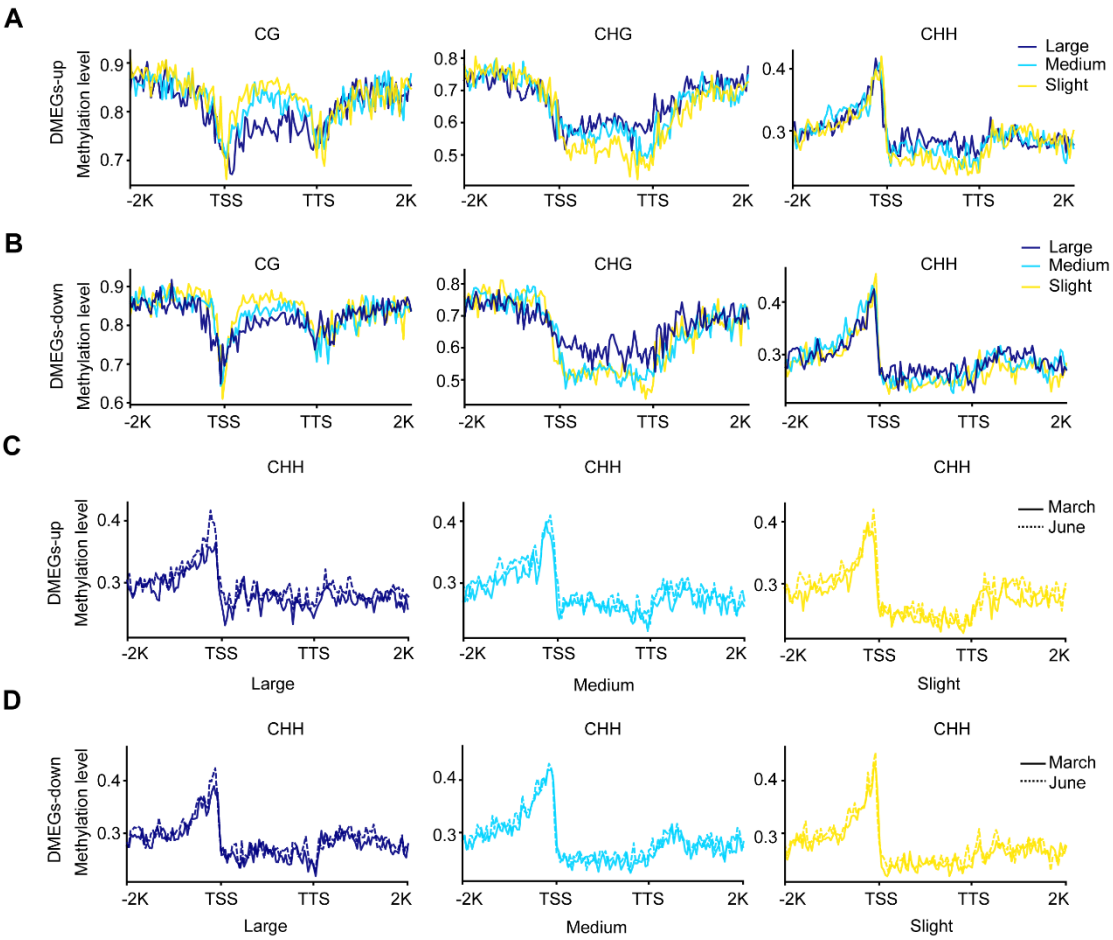


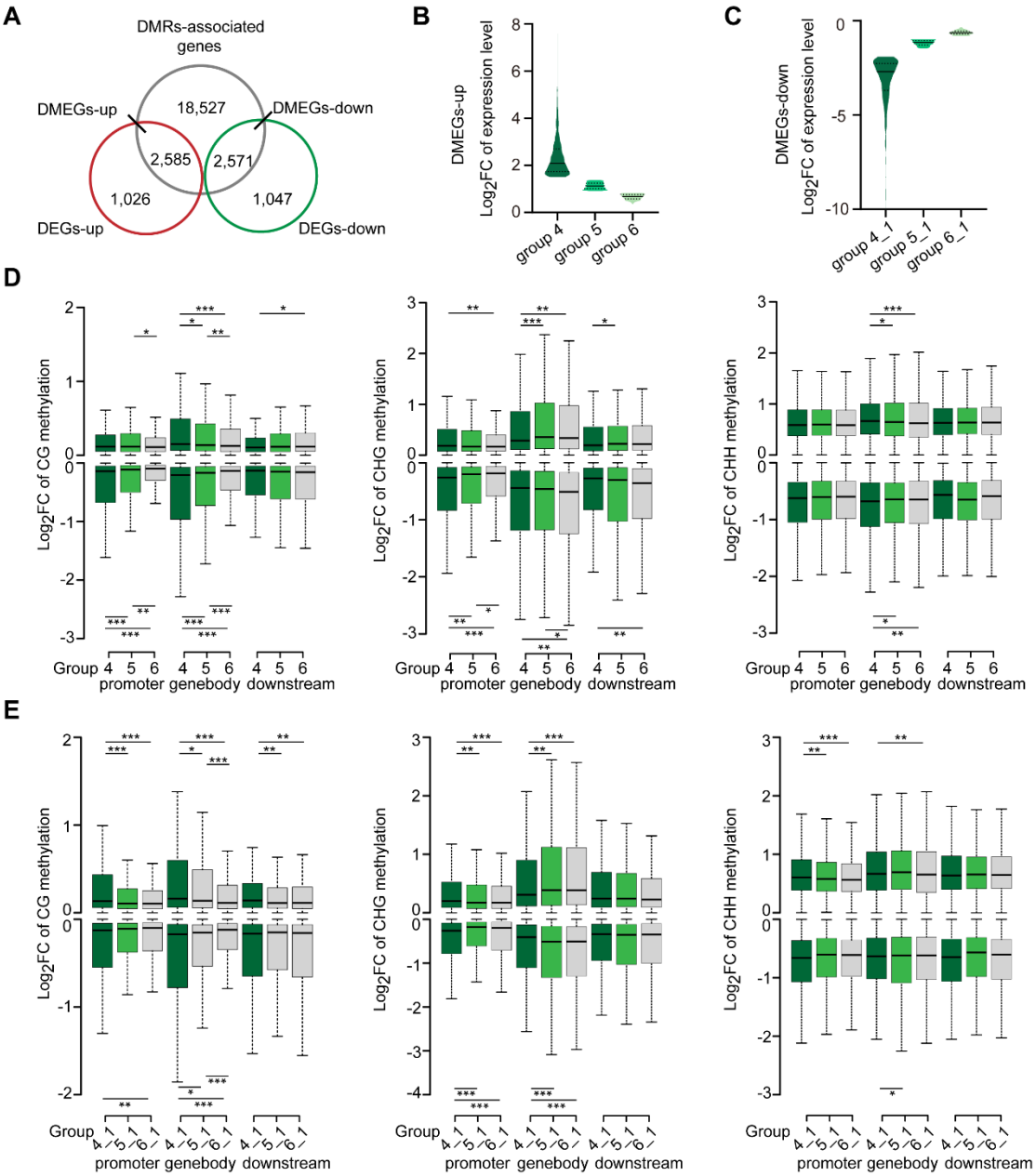
Supplementary Fig. 1. Statistical analysis of metabolome between March and June new shoots. (A) Score scatter plot of OPLS-DA model for metabolome data of March and June new shoots. (B) Volcano plot of differentially accumulated metabolites (DAMs). (C-D) KEGG enrichment analyses on the up- and down-regulated DAMs.



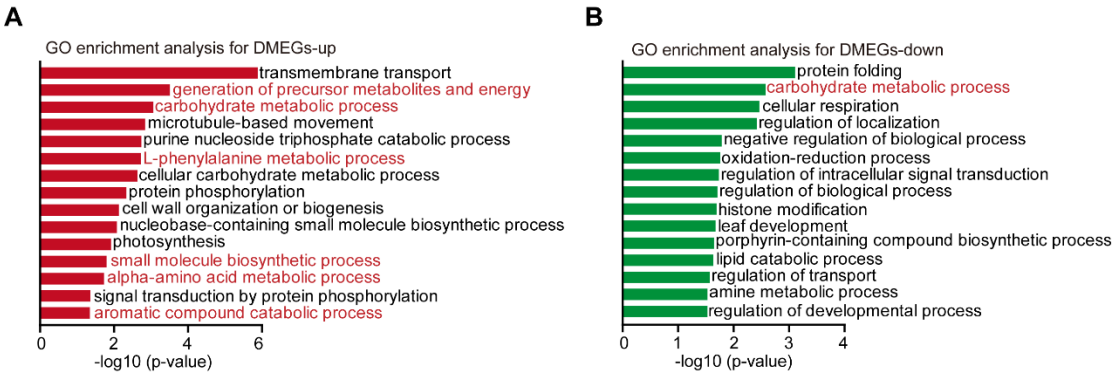
Supplementary Fig. 2. Phylogenetic analysis of DNA methyltransferase, demethylase encoding genes and genes involved in 5mC methylation in tea plant genome. The red five-pointed stars represent differentially expressed genes.



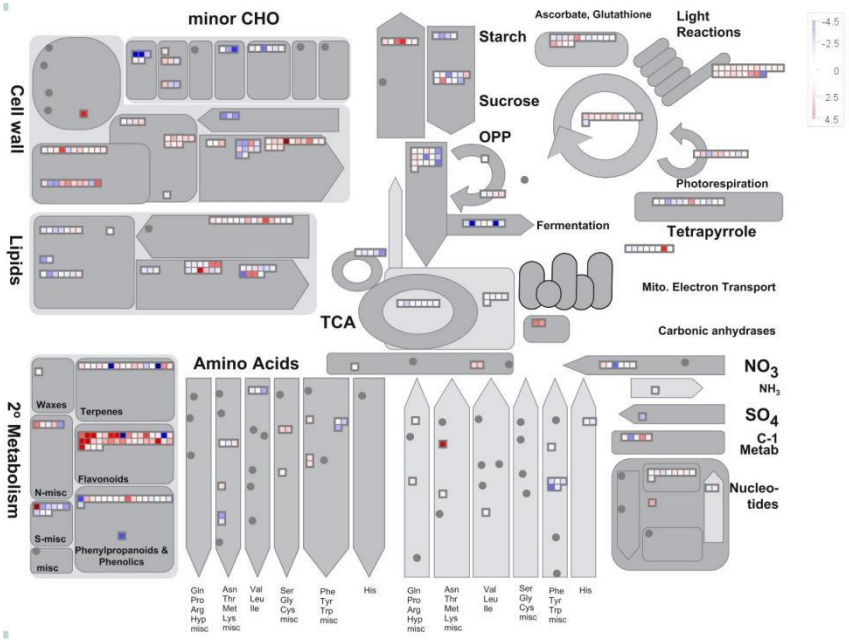
Supplementary Fig. 3. 5mC methylation levels of different gene groups and changes of CHH methylation levels. (A-B) The 5mC methylation levels of DMRs-associated DEGs (DMEGs-up, DMEGs-down) in largely, moderately and slightly changed groups. DMEGs-up and DMEGs-down grouping was shown is Fig. 4B, C. (C-D) CHH methylation levels of DMEGs-up and DMEGs-down in largely, moderately and slightly changed gene groups in March and June new shoots.



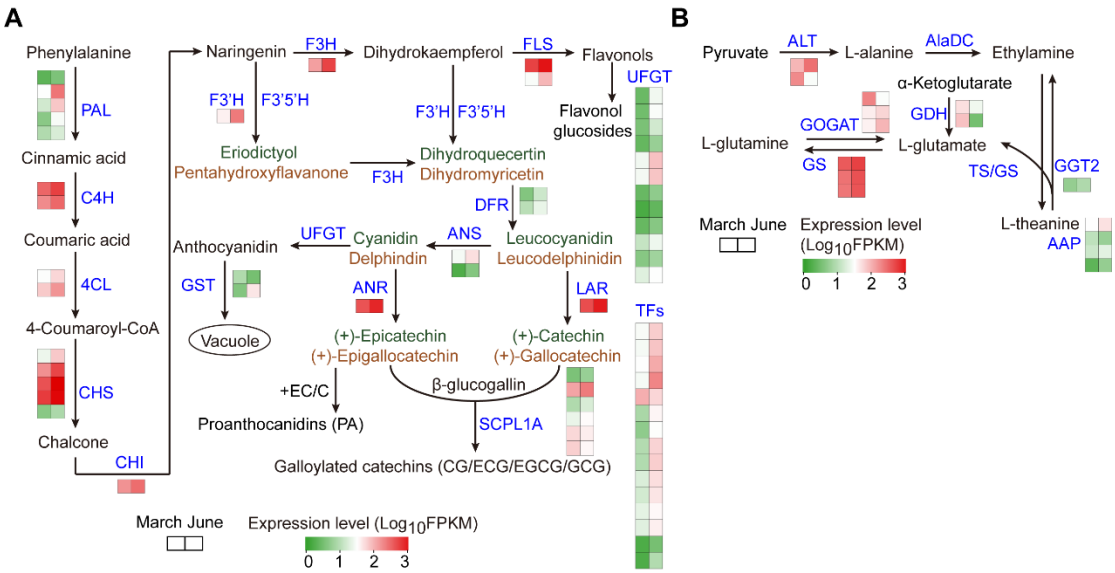
Supplementary Fig. 4. Log_2FC of 5mC methylation levels of differently expressed gene groups that divided according to the changes of expression levels. (A) Number of differently expressed DMRs-associated genes. (B-C) Log_2FC of expression levels of different groups for up- and down-regulated DMRs-associated DEGs, respectively. (D) Log_2FC of 5mC methylation levels of DEMGs-up. (E) Log_2FC of 5mC methylation levels of DEMGs-down**.** Wilcoxon rank sum tests: $*P \leq 0.05$; $**P \leq 0.01$; $***P \leq 0.001$.



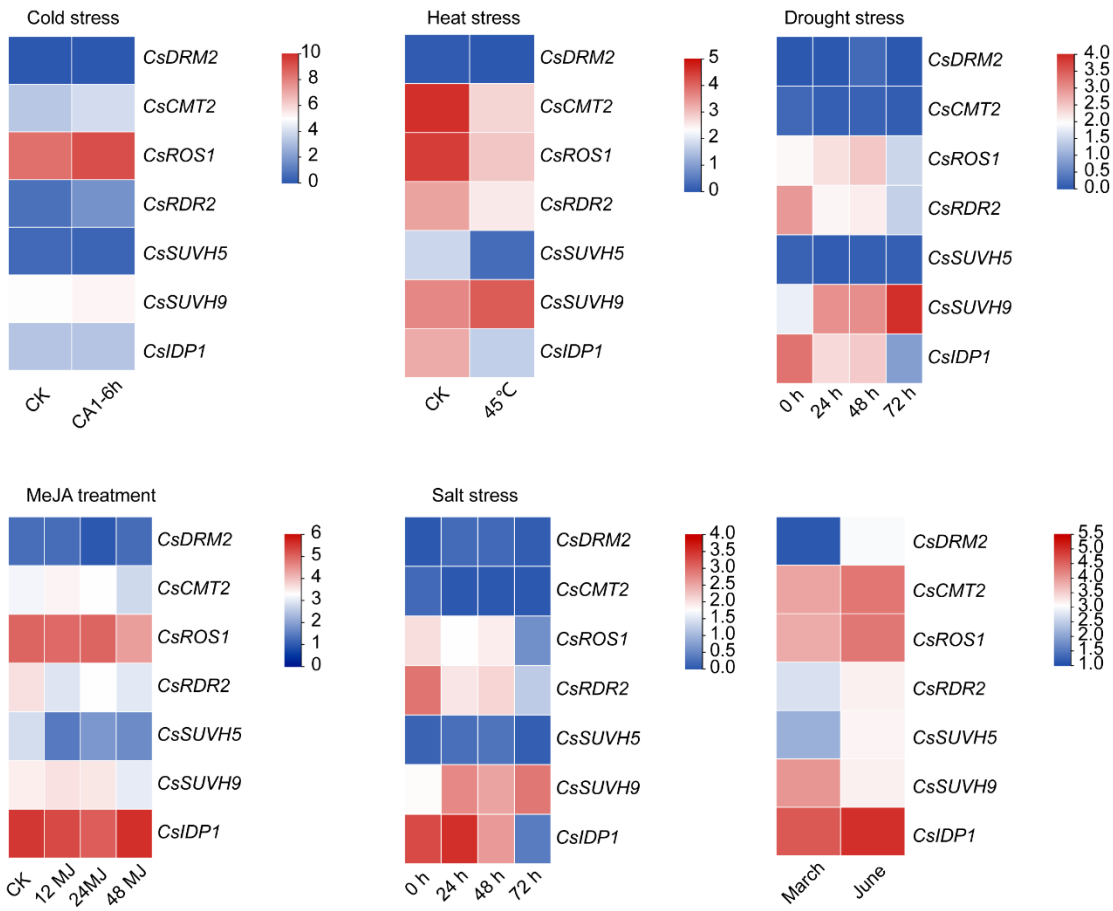
Supplementary Fig. 5. GO enrichment analysis of CG, CHG and CHH DMRs associated up- and down-regulated genes.



Supplementary Fig. 6. ManMap analysis of CG, CHG and CHH DMRs-associated DEGs.



Supplemental Fig. 7. The changes of gene expression in flavonoids and theanine synthesis pathway. (A) DEGs in flavonoids synthesis pathway. (B) DEGs in theanine synthesis pathway.



Supplemental Fig. 8. The expression levels of DNA methyltransferase, demethyltase encoding genes and genes involved 5mC methylation in the new shoots under different treatments.